SIMATS ENGINEERING

ASSESSMENT TOOL 3:

MCQ Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Total Marks: 30

Dave's Taxonomy: Precision (Level 3)
Question Count: 15
Duration : 60 Minutes

Code of Conduct

I Rakshana A.R. (192126002 7) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.R.

MCQ Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions

Total Marks Awarded:

1. A company cloud storage requires 5TB of data replication across three regions. When is the cloud storage Required?

a) 5TB
b) 10 TB
c) 15 TB
d) 20 TB

Ans: c) 15 TB

2. A Cloud Storage Provides offers at 99.99% uptime SLA. How much downtime can be expected per month?
a) 4 min, b) 40 min, c) 4 hr, d) 24 hrs Ans: 4 min

3. A cloud based load balance distributes 60,000 requests per sec among 10 servers what is the avg no. of request per server per sec?
a) 5000, b) 6000, c) 7000, d) 10000 Ans: B) 6000

4. A company cloud network has 1 GBPs band with & needs to transfer 500GB of data. how long will it take? Ans: c) 1.1
a) 1 hr, b) 10 hr, c) 1.1 hr, d) 5.5 hr

5. A cloud storage system experiances a 99.95% availability rate how much downtime does it experience a year?
a) 4 hr, b) 8 hr, c) 20 hr, d) 40 hr. Ans: A) 4 hr

6. A cloud Provider offers Pay-as-you-go pricing at $0.05 per GB for storage. If a company stores 10TB what is the monthly cost?
a) $50, b) $500, c) $1000, d) $2000. Ans: B)

7. A cloud network has 10 Gbps throughput but experiences 2% package loss. what is the effective throughput?
a) 9.8 Gbps, b) 9.6 Gbps, c) 8.0 Gbps, d) 7.5 Gbps. Ans: A) 9.8 Gbps

8. A user uploads 250 MB of data per day to the cloud. How much data will be uploaded in one year (365 days)?
a) 91.25 GB, b) 8.5 GB, c) 100 GB, d) 120 GB Ans: A) 91.25 GB

9. A cloud based video streaming service requires 5 Mbps bandwith per user. If 100 users are streamming simulatiously what is total required band with.

a) 500 Mbps, b) 250 Mbps, c) 200 Mbps, d) 1000 Mbps. Ans: A) 500 Mbps

10. A cloud Provides a Gaurenty 99.95 % optained per year. How much downtime does this allow in min / year?
a) 262 min b) 365 min, c) 525 min, d) 150 min. Ans: A) 262 min

11. A vishual data center has 200 physical servers if each server runs 40 VM. What is the total no. of VM in the data center.
a) 4,000, b) 8000, c) 6000, d) 10,000. Ans: B) 8000

12. A physical server has 128 GB of RAM & hosts 10 VM, each requiring 12 GB. How much memory is overcommitted?
a) 4 GB, b) 6 GB, c) 8 GB, d) 10 GB. Ans: c) 8 GB

13. A VM requires 4 Vcpus if a server has a total of 64 cores & an overcommitment ratio of 2:1, how many VM can be hosted?
a) 16, b) 32, c) 64, d) 128. Ans: B) 32

14. A Company migrates 50 on permise servers to the cloud reducing Power consuption by 60%. If the orginal overcommit Power usage was 100. what is new power consuption.
a) 40 KW, b) 50 KW, c) 60 KW, d) 70 KW Ans: c) 60 KW

15. A cloud provider operates a virtualized data center with 100 physical servers, each with 64 VCPU. If the overcommitment ratio 4:1 how many total VCPUs can be allocated.
a) 6400, b) 12,500, c) 25,600, d) 50,000
Ans: c) 25,600

16. which cloud deployment model is used for organization that reference both private & Public cloud serveree?
a) Hybrid clould, b) public cloud, c) Private cloud, d) Community cloud.

Ans: A) Hybrid Cloud.

17. which Cloud Seavrity stategy ensures that sensitive data remains private even if interepted.
a) Encryption, b) Load balancing, c) Virtualization, d) Resource Pooling.

Ans: (A) encryption

18. If a hypervisor losts 50 VMs & each VM use 20 GB bandwith per day calculate the total bandwith consumed per week.
a) 5000 GB, b) 7000 GB, c) 6000 GB, d) 7500 GB

Ans: (B) 7000 GB

19. A cloud system runs 76% avg CPU utilization. If has 32 cores at 2.5 hrs each total utilized processing power?
a) 5.6 GHz, b) 5.8 GHz, c) 64 GHz, d) 70 GHz. Ans: (A) 5.6 GHz

20. If network, latency in a virtualized system is 5ms & processing time is 10ms Calculate total round trip delay for 100 requests.
a) 1.5 sec, b) 2 sec, c) 1 sec, d) 3 sec

Ans: (A) 1.5 Sec